

Bridging the Unconnected: A Synthesis of Information Geometry, Fractal State-Space Dynamics, and Non-Perturbative Renormalization

Executive Summary & Taxonomy Matrix

This report presents a rigorous academic literature survey investigating the intersections between Bohmian Mechanics (BM), Invariant Set Theory (IST), and Asymptotic Safety (AS). The analysis was conducted across formal physics repositories including arXiv, INSPIRE-HEP, and the SAO/NASA Astrophysics Data System, adhering to a stratified two-tiered timeline. Tier 1 includes foundational works establishing the bedrock of each pillar, such as Bohm (1952) for BM, Palmer (2009/2016) for IST, and Reuter (1998) for AS. Tier 2 focuses on modern technical advances from 2020–2026 to extract high-precision machinery.

The central finding is that while no formal literature explicitly unifies all three pillars—BM, IST, and AS—the landscape reveals a rich set of "near misses," powerful implicit structural homologies, and critical adjacent domains containing the necessary components for a synthetic framework. The most significant intersection is found not within direct publications combining all three theories, but in the confluence of three distinct fields: Information Geometry, Fractal and p-Adic Geometries, and Koopman-von Neumann Mechanics. These adjacent domains provide the precise mathematical tools required to bridge the conceptual and ontological gaps between the three pillars. For instance, Information Geometry offers a first-principles derivation of the Bohmian quantum potential from Fisher information, directly addressing a key ontological debt ⁸⁹ ¹⁹⁶. Similarly, p-adic and non-commutative geometries offer a rigorous mathematical language to formalize IST's non-computable, fractal state-space manifold and potentially tame UV infinities in quantum gravity ³⁸ ⁹². Finally, the KvN formalism provides a mechanism for emergent linear unitary evolution from underlying deterministic dynamics, a crucial feature for bridging classical determinism and quantum linearity ¹⁰¹

The survey classifies the maturity of each domain intersection. The combination of Bohmian Mechanics and Asymptotic Safety represents a completely unexplored territory, with no known precedents applying pilot-wave trajectories to fRG flows ^{72 217}. The pairing of Invariant Set Theory and Asymptotic Safety is characterized by sparse precedents and strong implicit parallels, primarily centered on the concept of "fractality" appearing in both cosmological state-space geometry and Planck-scale spacetime structure ¹⁸⁷²⁵². The intersection of Invariant Set Theory and Bohmian Mechanics is the most developed among the pairs, though still sparse, with connections rooted in their shared rejection of statistical independence and their commitment to definite particle trajectories ^{12 13 328}. However, a concrete mathematical formulation linking IST's fractal invariant set to the dynamics of the Bohmian quantum potential remains an open problem.

Based on this analysis, the primary task for the v6.3 unified framework is not to invent new mathematics but to act as an integrator, assembling these pre-existing, powerful tools into a coherent whole. The major barriers identified are conceptual discontinuity (bridging deterministic ontology with standard probabilistic QFT), mathematical complexity, and ensuring empirical testability. This report concludes by providing a detailed taxonomy matrix mapping the current state of research and a feature extraction matrix outlining actionable concepts from adjacent fields to strengthen the v6.3 framework's internal consistency, synergy, resilience, and elegance.

Domain Pairing / Triad	Maturity Level	Primary Contributors / Foundational Works
Bohmian Mechanics + Asymptotic Safety (BM + AS)	Completely Unexplored Territory (Gap)	No known formal publications found connecting these two frameworks.
Invariant Set Theory + Bohmian Mechanics (IST + BM)	Sparse Precedents	Tim Palmer (IST) ¹⁸⁷ , David Bohm (BM) ⁶⁵ , Antony Valentini, Justin Barrett. Conceptual links via superdeterminism and counterfactual reasoning ^{12 13 15} .
Invariant Set Theory + Asymptotic Safety (IST + AS)	Sparse Precedents / Implicit Parallels	Tim Palmer (IST) ²⁸⁸ , Martin Reuter (AS) ⁷ . Shared use of "fractality" ¹¹⁶²⁵² and p-adic numbers ²⁶⁸ as potential mathematical bridges ³⁸ .
Bohmian Mechanics + IST + AS (Triad)	Completely Unexplored Territory (Gap)	No known formal publications found unifying all three frameworks.

Detailed Literature Analysis by Domain Intersection

Pairwise Domain A: Bohmian Mechanics and Asymptotic Safety (BM + AS)

The search for explicit academic literature formally combining Bohmian Mechanics (BM) with Asymptotic Safety (AS) yielded no direct publications. There are no known studies that apply pilot-wave theory, either for particles or fields, to the context of quantum gravity within the AS program. Specifically, no research has been found that investigates the application of pilot-wave trajectories to the Functional Renormalization Group (fRG) flow of gravitational couplings, nor has any work been identified that proposes a scale-dependent guidance equation or a running quantum potential (Q_k) evolving under the RG flow. Similarly, there are no published attempts to construct a pilot-wave extension of the non-trivial ultraviolet (UV) fixed point that defines the AS scenario, nor to define the bare action of AS using a pilot-wave ontology. This complete absence of precedent establishes the BM+AS intersection as a *Completely Unexplored Territory (Gap)*.

Despite the lack of explicit unifications, several implicit parallels and tantalizing hints suggest a potential for future synthesis. Both BM and AS operate within the broader framework of quantum field theory (QFT) and rely heavily on the concept of the renormalization group (RG) flow. Asymptotic Safety posits that a consistent and predictive quantum theory of gravity exists because its coupling constants run under the RG flow to approach a non-Gaussian UV fixed point ¹³²¹³³²⁴¹. This fixed point ensures that physical quantities remain finite and well-behaved at arbitrarily high energies ²⁰⁹. In parallel, some interpretations of Bohmian mechanics see the wavefunction itself as being defined on the unstable manifold of a fixed point action ²³². The RG flow connects the UV regime, where the effective action Γ_k approaches a bare (fixed point) action S , to the infrared (IR) limit where all fluctuations are integrated out ⁷². This suggests a deep connection between the geometric structure of the theory space in AS and the dynamical state of the system in BM.

One source explicitly mentions identifying the RG cutoff scale k with the Planck scale M_{Pl} , which would create a direct link between the scales of AS and the fundamental scale of quantum gravity ²¹⁷. This idea is further supported by the suggestion that the RG flow could solve problems like the cosmological constant problem, which is also a key motivation for theories like AS ⁶⁴. Furthermore, the existence of a geometric origin for the Bohm potential in the context of quantum gravity is noted, indicating that researchers perceive a conceptual link, even if it has not been formalized ⁶⁷. Another study explores the connection between the quantum potential and quantum fluctuations, which are

precisely the objects studied in non-perturbative methods like fRG ¹⁹⁶. The AS program uses non-perturbative methods, such as the Wetterich equation, to compute the full momentum dependence of propagators in quantum gravity ²¹²³⁵⁰. It is conceivable that a similar approach could be used to derive a scale-dependent quantum potential, although no such attempt has been documented.

A significant conceptual barrier to a BM+AS synthesis is the nature of time. Standard formulations of BM often rely on a preferred foliation of spacetime, which conflicts with the manifest Lorentz invariance (or general covariance) sought in quantum gravity. While recent work has produced Lorentz-covariant Bohmian trajectories, this remains an active area of research ²²³. AS, as a continuum QFT approach, also faces challenges in maintaining manifest background independence throughout the RG flow ^{3 214}. Any successful synthesis would need to reconcile these issues, perhaps by deriving a preferred time from the dynamics of the fRG flow itself. Another barrier is the ontological shift required. BM provides a clear ontology of particles with definite positions guided by a universal wavefunction ¹⁹⁰²⁹⁷. AS, however, operates primarily in the language of effective actions and coupling constants in theory space, lacking a built-in particle ontology. Connecting the two would require showing how the emergence of a stable, relational stationary state from the coarse-graining and decoherence of an ontic wavefunctional could give rise to the fixed-point attractor in AS ²³².

Pairwise Domain B: Invariant Set Theory and Bohmian Mechanics (IST + BM)

The intersection of Invariant Set Theory (IST) and Bohmian Mechanics (BM) is characterized by *Sparse Precedents*. There is no single publication that presents a fully worked-out mathematical model unifying the two frameworks. However, a number of sources explicitly mention the two theories in close proximity, suggesting a recognized conceptual kinship ^{12 13 15}. The connection is primarily philosophical and based on their shared ontological commitments rather than a deep mathematical formulation. The most prominent common ground is their stance on superdeterminism and the violation of statistical independence, a cornerstone assumption in Bell's theorem ⁵⁰. Bell's theorem implies that any local hidden variable theory must violate statistical independence, meaning the experimenter's choice of measurement settings is correlated with the properties of the particle being measured ⁵⁰.

IST provides a geometric mechanism for this correlation. It postulates the Invariant Set Postulate, which states that the true states of the universe correspond to a non-computable fractal set IU in cosmological state space ¹⁸⁷²⁵⁴. Physical reality consists only

of those cosmological states that lie on this set. Counterfactual states, which are necessary for Bell's argument, are simply not part of physical reality; they correspond to points in state space that are not on IU . This avoids any need for ad-hoc conspiracies, as the correlations are a geometric property of the universe's state-space geometry ²⁶⁸. Several authors note that this approach is philosophically similar to Bohmian mechanics, as both reject the notion of free, independent choices in experiments ^{12 15}. Indeed, one source explicitly places IST alongside Bohmian mechanics as examples of superdeterministic theories ¹³.

Both IST and BM share a commitment to a deterministic, trajectory-based ontology. IST posits that the state of the universe evolves deterministically on the fractal invariant set IU ¹⁸⁷. BM posits that particles have definite positions and trajectories at all times, guided by the wavefunction ¹⁸⁸²⁹⁷. The challenge for a formal IST+BM synthesis is to connect these two pictures of reality. How do the trajectories of BM evolve on the highly irregular, non-Euclidean geometry of IU ? How does the dynamics of the guiding wavefunction ψ relate to the geometry of IU ? Some preliminary steps exist; for example, Palmer has explored the role of complex numbers within IST, which is a step toward connecting its geometric postulate to the formalism of quantum mechanics ²⁹⁰. However, a concrete mathematical framework is missing.

Another point of contact is the equilibrium hypothesis. In BM, the Born rule ($\rho=|\psi|^2$) is typically treated as an additional assumption about the initial distribution of particle positions. It can be shown to be a stable equilibrium state under the Bohmian dynamics. The situation in IST is analogous; since the theory assumes that all physically possible states lie on IU , the distribution of states is inherently constrained. One author notes that as long as this "equilibrium hypothesis is fulfilled," the predictions of IST will align with those of standard quantum mechanics, much like in Bohmian mechanics ¹². This suggests that IST could potentially provide a deeper justification for the Born rule, grounding it in the measure-theoretic properties of the invariant set.

The primary barrier to progress is the immense mathematical difficulty of working with a non-computable, fractal manifold. While the Cantor set topology is a starting point, developing a calculus and differential geometry on such a set is a profound challenge ²⁶⁹. Without this, it is impossible to write down a meaningful guidance equation for particle trajectories. Furthermore, the relationship between the ontic wavefunctional of BM and the geometry of IU is entirely unknown. Does the wavefunctional encode the shape of IU ? Does it act as a force field on the trajectories? These questions represent the core of the research gap. The most promising path forward may involve borrowing tools from

adjacent fields, such as p-adic analysis, which provides a natural mathematical setting for dealing with non-Archimedean, fractal-like structures 52 268.

Pairwise Domain C: Invariant Set Theory and Asymptotic Safety (IST + AS)

The intersection of Invariant Set Theory (IST) and Asymptotic Safety (AS) is best described as having *Sparse Precedents and Strong Implicit Parallels*. There is no known formal literature that attempts to merge IST's cosmological state-space geometry with the spacetime geometry of AS. However, the two frameworks share a strikingly similar use of the term "fractality," pointing towards a deep structural homology that has not yet been explored. AS predicts that spacetime at the Planck scale possesses a fractal structure, with its spectral dimension decreasing at high energies 118252. This means that the effective geometry of spacetime becomes increasingly irregular and non-classical at short distances 362. IST, conversely, posits that the entire state-space of the cosmos is a compact, non-computable fractal, specifically a Cantor set 288289398. While AS describes the geometry of spacetime, IST describes the geometry of the space of all possible cosmological histories.

This parallel suggests a potential, albeit speculative, connection. One could hypothesize that the fractal geometry of the AS spacetime manifold is not an accident but is instead encoded by the more fundamental, global geometry of IST's invariant set $I\mathcal{U}$. In this view, the local, scale-dependent properties of spacetime seen in AS would emerge from the global, topological properties of $I\mathcal{U}$. The challenge lies in making this connection mathematically precise. It requires moving from a state-space fractal to a spacetime fractal and finding a principled way to relate their respective geometries. The topology of a non-computable Cantor set must be reconciled with the continuous, albeit highly singular, geometry of a fractal spacetime.

A potential mathematical bridge between these concepts is provided by p-adic number theory, which is explicitly used in the foundations of IST 187268. P-adic numbers form a non-Archimedean field, meaning they obey the ultrametric triangle inequality ($|x-z|_p \leq \max(|x-y|_p, |y-z|_p)$). This leads to a very different geometric structure than the familiar Euclidean space, one that is inherently hierarchical and fractal-like. P-adic methods have been successfully applied to construct interacting Quantum Field Theories (QFTs) and explore their properties, demonstrating the viability of this mathematical framework for fundamental physics 92 253260. It is conceivable that the p-adic geometry used to define the state space in IST could be related to the non-perturbative, scale-dependent geometry of spacetime in AS. Recent work in arithmetic quantum field theory and the Adelic Physics Programme has begun to explore the role of non-Archimedean

geometry in physics, lending credence to this idea ³⁷¹³⁷³³⁷⁴. For example, the use of Bruhat-Tits trees in p-adic CFT provides a discrete, tree-like structure that is analogous to the causal sets or spin foams in other quantum gravity approaches ²⁷⁹.

The main barrier to an IST+AS synthesis is the vast difference in scope and scale. IST is a theory of the entire cosmos, linking quantum phenomena to cosmology in a way that is absent in standard physics ²⁸⁸³²². AS is a QFT-based approach focused on defining a consistent quantum theory of the gravitational field at the level of the Lagrangian or effective action ^{94 209}. Bridging these two perspectives requires a monumental leap in conceptual understanding. For example, the RG flow in AS changes the effective dimensionality and topology of spacetime as one moves from the UV to the IR ³⁹². How does this dynamic process interact with the static-but-fundamental geometry of IST's invariant set? Another issue is the status of background independence. AS aims for a diffeomorphism-invariant theory ³, while IST's state-space is defined relative to the global structure of the universe. Reconciling these two notions of background independence is a formidable task. Despite these challenges, the shared language of fractality and the availability of p-adic mathematics as a common tool make this an intriguing, if nascent, area for future research.

Triadic Domain: Bohmian Mechanics + Invariant Set Theory + Asymptotic Safety

The triad of Bohmian Mechanics (BM), Invariant Set Theory (IST), and Asymptotic Safety (AS) represents a *Completely Unexplored Territory (Gap)*. The literature survey confirms that no formal work has been published attempting to combine these three theoretical pillars into a single, coherent framework. There are no known papers that propose a model where deterministic Bohmian trajectories evolve on a non-computable, p-adic fractal invariant set IU , whose geometry is linked to the UV completion of gravity via an asymptotically safe fixed point. This complete absence of precedent highlights the novelty and potential impact of such a synthetic endeavor.

However, the analysis of the pairwise intersections and adjacent domains reveals a clear and compelling pathway toward constructing such a framework. The necessary components are scattered across several distinct fields of theoretical physics and mathematics, and the task for a v6.3 unified program is to assemble them into a cohesive structure. The proposed synthesis would rest on three pillars, each drawing on the most promising adjacent domains identified in the survey.

The first pillar is **Information Geometry**, which provides the mechanism to generate the quantum potential Q from first principles. The Hall-Reginatto formalism demonstrates that by assuming the conservation of Fisher information, one can derive the exact mathematical form of the Bohmian quantum potential, $Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}$ ^{89 196}. This elevates Q from a postulated term in the Schrödinger equation to a consequence of a fundamental principle of inference and information conservation. This approach directly addresses a key ontological question in any deterministic theory: why does the quantum potential exist? By linking it to the Cramér-Rao bound, it also provides a geometric foundation for the uncertainty principle itself ^{76 77 153}. This pillar would provide the crucial link between the statistical description of BM and the underlying deterministic particle dynamics.

The second pillar is **Fractal and p-Adic Geometry**, which offers a rigorous mathematical foundation for IST's ontological claims. The non-computable, fractal nature of IST's invariant set IU is difficult to handle with standard analysis. P-adic numbers and non-Archimedean geometry provide the ideal tools for this task ^{38 52}. These frameworks are designed to describe spaces that are granular, disconnected, and possess self-similar, hierarchical structures, all of which are characteristic of fractals ²⁶⁰. The development of rigorous p-adic QFTs shows that it is possible to perform quantum field theory over such exotic spaces ^{92 253}. This pillar would allow the v6.3 framework to move beyond the qualitative description of IU and begin to formulate a quantitative theory of physics on a fractal state-space. This approach could also offer a novel regularization scheme for quantum gravity, potentially taming trans-Planckian infinities in a way that respects background independence ¹³⁸⁴²¹.

The third pillar is **Koopman-von Neumann (KvN) Mechanics**, which provides the bridge to emergent linear unitary dynamics. A major challenge for any deterministic interpretation of quantum mechanics is explaining why the observable world appears to follow the linear, unitary laws of the Schrödinger equation. The KvN formalism is uniquely suited to address this. It recasts deterministic classical mechanics as an operator theory on a Hilbert space, leading to a linear evolution equation analogous to the Schrödinger equation ¹⁰¹¹⁰⁴. For chaotic systems, the spectrum of the Koopman operator can contain pure point, absolutely continuous, and singular continuous parts ³⁵⁶. The emergence of regular, quasi-periodic motion corresponds to the appearance of a pure point spectrum ¹⁶⁶. This provides a potential mechanism for how the macroscopic, averaged behavior of the deterministic, nonlinear system of IST could appear linear and unitary, thus resolving the apparent conflict between determinism and quantum linearity.

By integrating these three pillars—a geometric origin for the quantum potential, a rigorous foundation for a fractal state-space, and a mechanism for emergent linearity—the v6.3 framework could pioneer a novel synthesis of these three disparate areas of physics, creating a unified theory that is coherent, robust, and compelling.

Technical Gap & Barrier Mapping

A thorough analysis of the existing literature reveals several significant technical gaps and conceptual barriers that stand in the way of synthesizing Bohmian Mechanics (BM), Invariant Set Theory (IST), and Asymptotic Safety (AS). These friction points represent the most challenging aspects of building a unified framework and highlight the specific areas where innovation is required. The barriers span mathematical complexity, conceptual discontinuity, and practical implementation challenges.

One of the most profound barriers lies in the transition from microscopic deterministic chaos to macroscopic linear quantum dynamics. IST posits a fundamentally deterministic universe whose state evolves on a fractal invariant set ¹⁸⁷. Such systems are generically expected to exhibit chaotic behavior, with a complex and perhaps singular continuous spectrum ³⁵⁶. However, the observable world, as described by quantum mechanics, exhibits stable, quasi-periodic behavior corresponding to a pure point spectrum. The Koopman-von Neumann (KvN) formalism provides a mathematical framework to analyze this transition by representing classical dynamics on a Hilbert space ¹⁰¹²⁸⁵. The key challenge is to determine the conditions under which the dynamics on IST's invariant set IU would lead to a pure point spectrum, thereby giving rise to the stable orbits and regular oscillations that we observe. While some results exist for specific types of dynamical systems, a general proof or characterization for a non-computable, fractal set is currently unavailable. This "spectral chaos vs. pure point spectrum" problem is a major load-bearing friction point that must be resolved to establish coherence between the micro-dynamics and macro-observables ¹⁶⁶³⁵⁶.

A second major barrier is achieving background independence within a pilot-wave field theory. Asymptotic Safety is a continuum QFT-based approach that seeks to define a non-perturbatively renormalizable quantum theory of gravity, a goal that is intrinsically tied to background independence ^{3 214}. Most extensions of Bohmian mechanics to quantum field theory (QFT) are formulated in a fixed, Minkowski spacetime background ³⁶⁹⁴⁰⁴. While progress has been made in developing Lorentz-covariant versions of BM, achieving full general covariance and background independence remains a formidable technical

challenge ²²³. The unified framework would need to develop a pilot-wave theory for a gravitational field that is itself subject to an fRG flow controlled by a UV fixed point. This requires defining particle trajectories in a dynamical spacetime, which complicates the very notion of a configuration space on which the wavefunctional lives. The problem of topology change across an RG flow in AS adds another layer of complexity, as the framework must account for how the fixed-but-fundamental geometry of IST's invariant set interacts with a dynamically changing spacetime topology ³⁹².

Third, any application of the functional renormalization group (fRG) from Asymptotic Safety to a deterministic theory faces the problem of scheme dependence. The fRG equation is an integro-differential equation that depends on the choice of regulator function and the truncation scheme used to approximate the effective average action Γ_k ^{34 271}. Different choices can lead to different results, and ensuring that physical predictions are scheme-independent is a critical requirement for any reliable calculation ^{35 215}. A unified framework that uses the fRG to determine the parameters of a Bohmian guidance equation or a quantum potential would need to demonstrate that its core predictions are robust against these choices. This is a non-trivial task, as it requires proving that the emergent quantum behavior is a universal feature of the theory space, not an artifact of a particular computational method.

Finally, a conceptual barrier arises from the differing ontological commitments of the three frameworks. BM and IST are realist, trajectory-based theories that posit the existence of definite physical entities (particles, states) ¹⁸⁷²⁹⁷. AS, in its standard formulation, is an effective QFT framework that operates primarily in the language of generating functionals and coupling constants in theory space ³⁵³⁴⁰⁶. Bridging this gap requires showing how the ontic wavefunctional of BM, which guides the trajectories, emerges from the RG flow in AS. One might speculate that the UV fixed point in AS corresponds to a special, stable state of the ontic wavefunctional, but this connection is purely heuristic at present ²³². Furthermore, both BM and IST are superdeterministic theories that reject the statistical independence assumed in Bell-type arguments ^{13 50}. While this is a feature that allows them to evade Bell's theorem, it runs counter to the conventional wisdom of statistical physics and may make the theories seem less compelling to a wider audience unless their predictions are shown to be experimentally indistinguishable from standard quantum mechanics, except perhaps in extreme regimes. Overcoming these technical and conceptual barriers will require the integration of advanced tools from information geometry, non-Archimedean analysis, and ergodic theory, as suggested by the survey of adjacent domains.

Feature Extraction Matrix for Framework Optimization

To strengthen the v6.3 unified framework, this section presents a matrix of actionable concepts, theorems, and equations extracted from the reviewed literature. These features are drawn from adjacent, high-value domains and are categorized by the specific architectural debt in the v6.3 framework they are intended to address. Each entry specifies the feature, its mathematical or conceptual source, and its direct benefit to the goals of coherence, cohesiveness, robustness, and compellingness.

Feature / Concept	Source Context / Mathematical Tool	Benefit to v6.3
Derivation of Quantum Potential from Information Conservation	Hall-Reginatto information-geometric ensemble; variational principle on Fisher Information metric. 76 89 196	Coherence: Derives the Bohmian quantum potential Q from the fundamental principle of Fisher information conservation, avoiding its postulation as an ad-hoc term.
Geometric Origin of the Uncertainty Principle	Identification of mean quantum potential with Fisher information and its relation to the Cramér-Rao bound. 76 77 153	Robustness: Connects the Heisenberg uncertainty principle to information theory, strengthening the foundational underpinnings of the framework and linking it to a core principle of estimation theory.
Non-Computable State-Space Formalism	Tim Palmer's Invariant Set Postulate; Cantor set topology; p-adic integers as coordinates for state-space. 52 187254268	Cohesiveness: Provides a rigorous mathematical foundation for IST's ontological claim of a non-computable fractal state-space, enabling quantitative modeling.
Taming Trans-Planckian UV Infinities	P-adic Number Theory; Non-Archimedean Geometry; Ultrametric Triangle Inequality; Hierarchical Models. 38 92 260374421	Robustness: Offers a novel, non-perturbative regularization method for quantum gravity that is naturally immune to standard UV divergences arising from short-distance singularities.
Emergence of Linear Unitary Evolution	Koopman-von Neumann (KvN) Formalism; Spectral theory of dynamical systems; Odometers and Tame factors. 101166285332 356	Coherence: Provides a mechanism for how the linear, unitary evolution of the Schrödinger equation can emerge as an effective, macroscopic description of an underlying deterministic, nonlinear, and chaotic sub-quantum theory.
Connection Between Cosmology and Quantum Phenomena	Strong synergy between the global geometry of the cosmological state-space and local quantum laws. 144187288322	Compellingness: Explains the universality of quantum mechanics by rooting its applicability in the fundamental, large-scale geometric structure of the cosmos.
Information-Geometry to Spacetime Metric	Fisher Information Metric as a canonical Riemannian measure of local distinguishability, linked to spacetime geometry via holography. 301345349	Cohesiveness: Creates a direct, first-principles bridge between the information content of a physical system and the geometric structure of the spacetime it inhabits.
Explicit Construction of p-Adic QFTs	Rigorous construction of interacting QFTs over p-adic spaces; use of hierarchical models as analogues. 92 253260	Robustness: Validates the use of p-adic methods for fundamental physics, providing confidence that a quantum theory on a fractal manifold is mathematically viable.
Non-Perturbative Fixed Point in Theory Space	Functional Renormalization Group (fRG); Wetterich equation; Non-Gaussian UV fixed point in Asymptotic Safety. 7 72 132203	Cohesiveness: Provides the non-perturbative mechanism for the UV completion of the theory, ensuring predictivity at all energy scales without reliance on an expansion around a Gaussian fixed point.
Superdeterministic Evasion of Bell's Theorem	IST's Invariant Set Postulate; geometric constraint of counterfactuals; violation of statistical independence. 12 13 15 50	Robustness: Offers a principled, non-conspiratorial way to satisfy Bell's inequalities by denying the physical possibility of counterfactual measurement settings.
Quantum-Classical Limit as Flattening of Information Geometry	Fisher information metric characterizes the geometry of quantum states; the classical limit corresponds to a "flattening" of this geometry. 83 261	Cohesiveness: Provides a geometric criterion for the emergence of classical physics from quantum theory, linking it to the loss of quantum correlations and the flatness of the statistical manifold.

This matrix serves as a strategic guide for the v6.3 program, highlighting specific, high-impact concepts from the broader theoretical landscape that can be imported to resolve architectural weaknesses and build a more complete, compelling, and robust unified framework.

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